

TUTORIAL-1 DIGITAL SYSTEMS -EE1201

1 a. Convert the following numbers into decimal: (a) $(4310)_5$, (b) $(198)_{12}$, (c) $(EEE)_{16}$

b. Convert each of the following decimal numbers to unsigned binary, octal and hexadecimal numbers: (a) 27 (b) 915

2 a. Evaluate $(1011)_2 - (1101)_2$ using both 1's and 2's complement method.

b. Convert the following number systems: i. $(111101100)_2 = \text{GRAY code}$ ii. $(1101.101)_2 = ()_{10}$

3.

How many 1's are present in the binary representation of

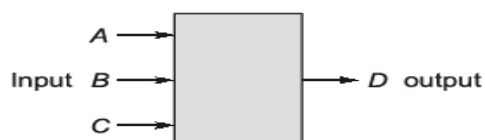
$$(4 \times 4096) + (9 \times 256) + (7 \times 16) + 5 ?$$

4.

Let $A = 11111010$ and $B = 00001111$ be two 8-bit 2's complement numbers. Their product in 2's complements is

5.

For the box shown the output D is true if and only if a majority of the inputs are true.



The Boolean function for the output is

6.

Let the Boolean function be defined as

$$f(A, B) = A' + B.$$

Simplify the Boolean expression $f(f(x + y, y), z)$ and obtain its final simplified form.

7. A bulb in a staircases has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by and one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles.

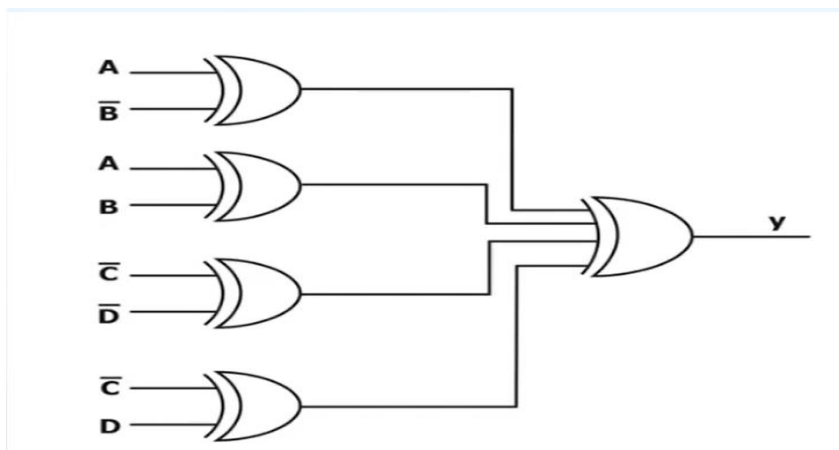
8 a. Simplify the following Boolean Expression to a minimum number of literals:

$$(BC' + A'D)(AB' + CD')$$

b. Simplify the given Boolean Expression $F = x' + xy + xz' + xy'z'$.

9 a. Implement $Y'Z + XY + X'Y'$ using only NOR gates.

b. What will be the output Y.

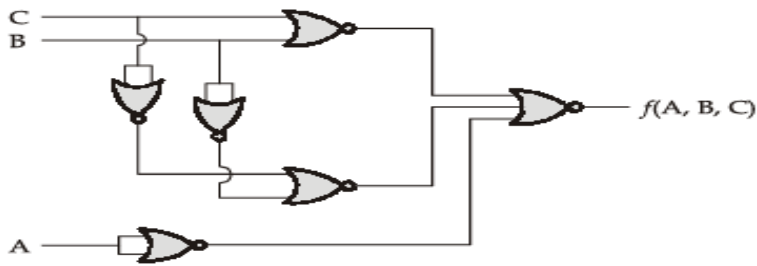


c. $f(A, B, C, D) = \pi M(1, 3, 8, 9, 10, 11, 14)$. d (7, 15)

Minimize and realise using only NOR gates.

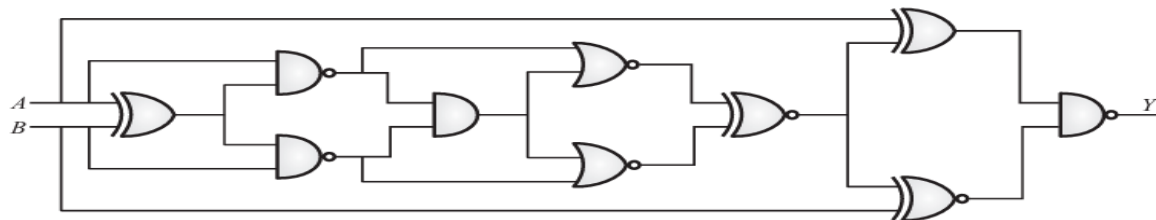
10. Give the expression for following output $f(A, B, C)$

A function $f(A, B, C)$ is produced by using NOR-gates as shown in the figure below:



11

Consider the logic circuit shown in figure below,



If $A \neq B$, then the expression for output 'Y' is,

12a.

2. Find the minimal POS of the given K-maps.

(a)

cd \ ab	00	01	11	10
00	1	1	1	1
01	x			
10	x			
11	1	1		x

(b) Wxyz

yz \ wx	00	01	11	10
00	0	x	0	x
01	x	1	x	1
10	0	x	1	0
11	0	1	x	0

(c)

cd \ ab	00	01	11	10
00	x	x	1	x
01	0	1	1	0
10		1	1	
11				

(d)

yz \ wx	00	01	11	10
00	1	1	d	d
01		d	1	
10	1	d		d
11				

(d → refers to don't care)

b. $f(A, B, C, D) = \sum m(1, 5, 7, 8, 10, 12, 13, 15)$ find total prime implicants and essential prime implicants.

13.

What is the minimal form of the Karnaugh map shown below? Assume that X denotes a don't care term.

<i>cd</i> \ <i>ab</i>		00	01	11	10
	00	1	X	X	1
	01	X			1
	11				
	10	1			X

14. Convert $Y = A + BC' + AB + A'BC$ into canonical sum of products(or SSOP) form.